

Curriculum Vitae

Inpyo Song

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EDUCATION

B.S., Major in Mechanical Engineering, Korea Aerospace University (KAU), Go-yang, Rep. Korea, Feb. 2023
<https://www.kau.ac.kr>

M.S., Immersive Media Engineering
Sungkyunkwan University (SKKU), Seoul, Rep. Korea
Sep. 2023 – Feb. 2026

- GPA: 3.79 / 4.0
- Intelligent and Interactive Systems Lab

Ph.D., Immersive Media Engineering
Sungkyunkwan University (SKKU), Seoul, Rep. Korea
Mar. 2026 – Present

- Intelligent and Interactive Systems Lab
- Advisor: Prof. Jangwon Lee

PROFESSIONAL EXPERIENCE

Indiana University Bloomington, Bloomington, IN, USA
<https://bloomington.iu.edu/>
Research Intern, Computer Vision Lab (<http://vision.soic.indiana.edu>)
Jun. 2024 – Aug. 2024

PUBLICATIONS

International Conference Papers

1. **Inpyo Song**, Jangwon Lee, “*Bounding-Box Trajectories Matter for Video Anomaly*”, European Conference on Computer Vision (ECCV), 2026. [accepted]
2. **Inpyo Song**, Jangwon Lee, “*Real-Time Traffic Accident Anticipation with Feature Reuse*”, IEEE International Conference on Image Processing (ICIP), Sep. 2025.
3. **Inpyo Song**, Hyemin Hwang, Jangwon Lee, “*PawPrint: Whose Footprints Are These? Identifying Animal Individuals by Their Footprints*”, IEEE International Conference on Image Processing (ICIP), Sep. 2025.
4. **Inpyo Song**, Sanghyeon Lee, Minjun Joo, Jangwon Lee, “*Anomaly Detection for People with Visual Impairments Using an Egocentric 360-Degree Camera*”, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Feb. 2025.
5. **Inpyo Song**, Jangwon Lee, “*SFTrack: A Robust Scale and Motion Adaptive Algorithm for Tracking Small and Fast Moving Objects*”, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Oct. 2024. **(Long Oral)**
6. **Inpyo Song**, Jongmin Lee, Moonwook Ryu, Jangwon Lee, “*Motion-Aware Heatmap Regression for Human Pose Estimation in Videos*”, International Joint Conferences on Artificial Intelligence (IJCAI), Aug. 2024.
7. **Inpyo Song**, Minjun Joo, Joonhyung Kwon, Jangwon Lee, “*Video Question Answering for People with Visual Impairments Using an Egocentric 360-Degree Camera*”, In IEEE Conference on Computer Vision and Pattern Recognition Workshop (CVPRW) on First Joint Egocentric Vision (EgoVis), Jun. 2024.
8. **Inpyo Song**, Jangwon Lee, “*Scale and Motion Adaptive Multi-Object Tracking Algorithm for Unmanned Aerial Vehicles*”, IEEE International Conference on Robot & Human Interactive Communication (RO-MAN), Late-Breaking Reports, Aug. 2023.

International Journal Papers

1. **Inpyo Song**, Moonwook Ryu, Jangwon Lee, “*Action-Conditioned Contrastive Learning for 3D Human Pose and Shape Estimation in Videos*”, Computer Vision and Image Understanding (CVIU), Sep. 2024. (IF: 4.3, **Q1**(2023))

Korean Conference Papers

1. **Inpyo Song**, Eunji Jeon, Jangwon Lee, “*Probing H.264 Compressed-Domain Signals for Video Anomaly Detection*”, The Korean Institute of Broadcast and Media Engineers (KIBME) Summer Conference, Jun. 2026.
2. Moonwook Ryu, Jongwoo Choi, **Inpyo Song**, Jangwon Lee, “*A Study on Synthetic Data Generation and Evaluation for Pose Estimation in Surveillance Videos*”, The Korean Institute of Broadcast and Media Engineers (KIBME) Summer Conference, Jun. 2026. (**Outstanding Poster Award**)
3. Jongmin Lee*, **Inpyo Song*** (Equal Contribution), Moonwook Ryu, Jangwon Lee, “*Smooth and Consistent Pose Estimation with Intersections Module*”, The Korean Institute of Broadcast and Media Engineers (KIBME) Summer Conference, Jun. 2024.
4. **Inpyo Song**, Jangwon Lee, “*Multi-task Learning in Facial Kinship Verification*”, The Korean Institute of Broadcast and Media Engineers (KIBME) Summer Conference, Jun. 2023.

Korean Journal Papers

1. **Inpyo Song**, Jangwon Lee, “*Teacher Response Entropy Based Adaptive Knowledge Distillation*”, Journal of Korean Institute of Intelligent Systems (KIIS), Dec. 2024.
2. **Inpyo Song**, Jangwon Lee, “*Rule-based Zero-shot Video Anomaly Detection Using Object Detection and Semantic Segmentation*”, Journal of Broadcast Engineering (JBE), Vol. 29, No. 6, Nov. 2024.

MANUSCRIPTS UNDER REVIEW

1. **Inpyo Song**, Jangwon Lee, “A VLM Answer is Not an Anomaly Score: Rank Compression in Training-Free Video Anomaly Detection”
2. **Inpyo Song**, Jangwon Lee, “Frame-level Evaluation in Weakly Supervised Video Anomaly Detection Mostly Measures Video-Level Ranking”
3. **Inpyo Song**, Jangwon Lee, “Rethinking Open-World Video Anomaly Detection: Diagnosing Definition Blindness”
4. **Inpyo Song**, Eunji Jeon, Jangwon Lee, “PCEval: A Benchmark for Evaluating Physical Computing Capabilities of Large Language Models” (submitted)
5. **Inpyo Song**, Joonhyung Kwon, Eunji Jeon, Minjun Ju, Jangwon Lee, “Instance-Aligned Captions for Explainable Video Anomaly Detection” (submitted)

PATENTS

Korean Patent Applications

1. Jangwon Lee, **Inpyo Song**, “*Apparatus for Predicting Posture and Shape of 3D Human and Method Thereof*”, Research & Business Foundation of Sungkyunkwan University, Korean Patent Application No. 10-2024-0039467, Mar. 2024.
2. Jangwon Lee, **Inpyo Song**, Minjun Ju, Joonhyung Kwon, “*Apparatus for Detecting Abnormality and Method Thereof*”, Research & Business Foundation of Sungkyunkwan University, Korean Patent Application No. 10-2024-0031055, Mar. 2024.
3. Jangwon Lee, Jongmin Lee, **Inpyo Song**, “*Apparatus for Estimating Human Pose and Method Thereof*”, Research & Business Foundation of Sungkyunkwan University, Korean Patent Application No. 10-2023-0186741, Oct. 2023.

International Patent Applications

1. Jangwon Lee, **Inpyo Song**, Minjun Ju, Junhyung Kwon, “*Apparatus for Detecting Abnormality and Method Thereof*”, Research & Business Foundation of Sungkyunkwan University, U.S. Patent Application No. 19/062,397, Feb. 2025.
2. Jangwon Lee, Jongmin Lee, **Inpyo Song**, “*Apparatus for Estimating Human Posture and Method Thereof*”, Research & Business Foundation of Sungkyunkwan University, U.S. Patent Application No. 18/979,379, Dec. 2024.

HONORS AND AWARDS

- **Outstanding Poster Award** (Poster Presenter), “*A Study on Synthetic Data Generation and Evaluation for Pose Estimation in Surveillance Videos*”, KIBME Summer Conference, Jun. 2026
- **Best Graduate Research Award, Inpyo Song**, “*Technological Social Responsibility: Anomaly Detection for People with Visual Impairments Using an Egocentric 360-Degree Camera*”, 2025 Digital Innovation Talent Symposium, Aug. 2025.
- **Excellence Award**, Jongmin Lee, **Inpyo Song**, Hannah Cho, “*Multimodal VQA-Based Tutoring System for the Digital Transformation of Hands-On Education*”, SKKU Graduate Student Startup Competition, Oct. 2024.
- **President’s Award from IITP** (Institute of Information & Communications Technology Planning & Evaluation), Jongmin Lee, **Inpyo Song**, “*Motion-Aware Heatmap Regression for Human Pose Estimation in Videos*”, 2024 Digital Innovation Talent Symposium, Aug. 2024.
- **Encouragement Award, Inpyo Song**, “*Technological Social Responsibility: Anomaly Detection for People with Visual Impairments Using an Egocentric 360-Degree Camera*”, SKKU Research Matters, Feb. 2024.
- **Grand Award**, Heetae Lee, **Inpyo Song**, Suho Jang, “*Generating Digital Twin using Instance-NeRF*”, KAU Start-up Idea Competition, Dec. 2022.

RESEARCH INTERESTS

Computer Vision : Anomaly Detection, 2D/3D Human Pose Estimation, Object Tracking

Deep Learning : Dataset Condensation, Knowledge Distillation

Multi-Modal : Visual Grounding, Visual Question Answering

SERVICE

International Conference Service

1. Association for the Advancement of Artificial Intelligence (AAAI), 2027, Reviewer
2. Conference on Neural Information Processing Systems (NeurIPS), 2026, Reviewer
3. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026, Reviewer
4. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026, Reviewer
5. Association for the Advancement of Artificial Intelligence (AAAI), 2026, Reviewer
6. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025, Reviewer
7. IEEE International Conference on Development and Learning (ICDL), 2024, Reviewer

REFERENCES

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